

5.4 List colouring

In this section, we take a look at a relatively recent generalization of the concepts of colouring studied so far. This generalization may seem a little far-fetched at first glance, but it turns out to supply a fundamental link between the classical (vertex and edge) chromatic numbers of a graph and its other invariants.

Suppose we are given a graph $G = (V, E)$, and for each vertex of G a list of colours permitted at that particular vertex: when can we colour G (in the usual sense) so that each vertex receives a colour from its list? More formally, let $(S_v)_{v \in V}$ be a family of sets. We call a vertex colouring c of G with $c(v) \in S_v$ for all $v \in V$ a colouring *from the lists* S_v . The graph G is called *k-list-colourable*, or *k-choosable*, if, for every family $(S_v)_{v \in V}$ with $|S_v| = k$ for all v , there is a vertex colouring of G from the lists S_v . The least integer k for which G is *k-choosable* is the *list-chromatic number*, or *choice number* $\text{ch}(G)$ of G .

List-colourings of edges are defined analogously. The least integer k such that G has an edge colouring from any family of lists of size k is the *list-chromatic index* $\text{ch}'(G)$ of G ; formally, we just set $\text{ch}'(G) := \text{ch}(L(G))$, where $L(G)$ is the line graph of G .

In principle, showing that a given graph is *k-choosable* is more difficult than proving it to be *k-colourable*: the latter is just the special case of the former where all lists are equal to $\{1, \dots, k\}$. Thus,

$$\text{ch}(G) \geq \chi(G) \quad \text{and} \quad \text{ch}'(G) \geq \chi'(G)$$

for all graphs G .

In spite of these inequalities, many of the known upper bounds for the chromatic number have turned out to be valid for the choice number, too. Examples for this phenomenon include Brooks's theorem and Proposition 5.2.2; in particular, graphs of large choice number still have subgraphs of large minimum degree. On the other hand, it is easy to construct graphs for which the two invariants are wide apart (Exercise 30). Taken together, these two facts indicate a little how far those general upper bounds on the chromatic number may be from the truth.

The following theorem shows that, in terms of its relationship to other graph invariants, the choice number differs fundamentally from the chromatic number. As mentioned before, there are 2-chromatic graphs of arbitrarily large minimum degree, e.g. the graphs $K_{n,n}$. The choice number, however, will be forced up by large values of invariants like δ , ε or κ :

Theorem 5.4.1. (Alon 1993)

There exists a function $f: \mathbb{N} \rightarrow \mathbb{N}$ such that, given any integer k , all graphs G with average degree $d(G) \geq f(k)$ satisfy $\text{ch}(G) \geq k$.

The proof of Theorem 5.4.1 uses probabilistic methods as introduced in Chapter 11.

Although statements of the form $\text{ch}(G) \leq k$ are formally stronger than the corresponding statement of $\chi(G) \leq k$, they can be easier to prove. A pretty example is the list version of the five colour theorem: every planar graph is 5-choosable. The proof of this does not use the five colour theorem (or even Euler's formula, on which the proof of the five colour theorem is based). We thus reobtain the five colour theorem as a corollary, with a very different proof.

Theorem 5.4.2. (Thomassen 1994)

Every planar graph is 5-choosable.

Proof. We shall prove the following assertion for all plane graphs G with at least 3 vertices: (4.2.8)

(*)

Suppose that every inner face of G is bounded by a triangle and its outer face by a cycle $C = v_1 \dots v_k v_1$. Suppose further that v_1 has already been coloured with the colour 1, and v_2 has been coloured 2. Suppose finally that with every other vertex of C a list of at least 3 colours is associated, and with every vertex of $G - C$ a list of at least 5 colours. Then the colouring of v_1 and v_2 can be extended to a colouring of G from the given lists.

Let us check first that (*) implies the assertion of the theorem. Let any plane graph be given, together with a list of 5 colours for each vertex. Add edges to this graph until it is a maximal plane graph G . By Proposition 4.2.8, G is a plane triangulation; let $v_1 v_2 v_3 v_1$ be the boundary of its outer face. We now colour v_1 and v_2 (differently) from their lists, and extend this colouring by (*) to a colouring of G from the lists given.

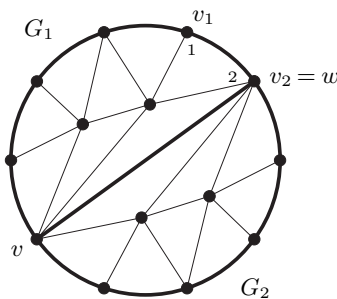


Fig. 5.4.1. The induction step with a chord vw ; here the case of $w = v_2$

Let us now prove (*), by induction on $|G|$. If $|G| = 3$, then $G = C$ and the assertion is trivial. Now let $|G| \geq 4$, and assume (*) for smaller graphs. If C has a chord vw , then vw lies on two unique cycles $C_1, C_2 \subseteq C + vw$ with $v_1v_2 \in C_1$ and $v_1v_2 \notin C_2$. For $i = 1, 2$, let G_i denote the subgraph of G induced by the vertices lying on C_i or in its inner face (Fig. 5.4.1). Applying the induction hypothesis first to G_1 and then – with the colours now assigned to v and w – to G_2 yields the desired colouring of G .

If C has no chord, let $v_1, u_1, \dots, u_m, v_{k-1}$ be the neighbours of v_k in their natural cyclic order around v_k ,¹ by definition of C , all those neighbours u_i lie in the inner face of C (Fig. 5.4.2). As the inner faces of C are bounded by triangles, $P := v_1u_1 \dots u_mv_{k-1}$ is a path in G , and $C' := P \cup (C - v_k)$ a cycle.

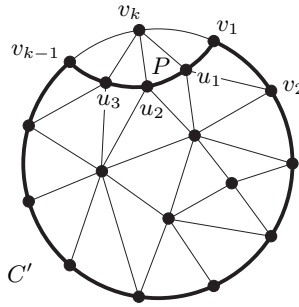


Fig. 5.4.2. The induction step without a chord

We now choose two different colours $j, \ell \neq 1$ from the list of v_k and delete these colours from the lists of all the vertices u_i . Then every list of a vertex on C' still has at least 3 colours, so by induction we may colour C' and its interior, i.e. the graph $G - v_k$. At least one of the two colours j, ℓ is not used for v_{k-1} , and we may assign that colour to v_k . $\square$

As is often the case with induction proofs, the key to the proof above lies in its delicately balanced strengthening of the assertion proved. Compared with ordinary colouring, the task of finding a suitable strengthening is helped greatly by the possibility to give different vertices lists of different lengths, and thus to tailor the colouring problem more fittingly to the structure of the graph. This suggests that maybe in other unsolved colouring problems too it might be of advantage to aim straight for their list version, i.e. to prove an assertion of the form $\text{ch}(G) \leq k$ instead of the formally weaker $\chi(G) \leq k$. Unfortunately, this approach fails for the four colour theorem: planar graphs are *not* in general 4-choosable.

¹ as in the first proof of the five colour theorem

As mentioned before, the chromatic number of a graph and its choice number may differ a lot. Surprisingly, however, no such examples are known for edge colourings. Indeed it has been conjectured that none exist:

List Colouring Conjecture. *Every graph G satisfies $\text{ch}'(G) = \chi'(G)$.*

We shall prove the list colouring conjecture for bipartite graphs. As a tool we shall use orientations of graphs, defined in Chapter 1.10. If D is a directed graph and $v \in V(D)$, we denote by $N^+(v)$ the set, and by $d^+(v)$ the number, of vertices w such that D contains an edge directed from v to w .

$N^+(v)$
 $d^+(v)$

To see how orientations come into play in the context of colouring, recall the greedy algorithm from Section 5.2. This colours the vertices of a graph G in turn, following a previously fixed ordering $(v_1, \dots, v_n)$, with the smallest available colour. This ordering defines an orientation of G if we orient every edge $v_i v_j$ ‘backwards’, that is, from v_j to v_i if $i < j$. Then to determine a colour for v_j the algorithm only looks at previously coloured neighbours of v_j , those to which v_j sends a directed edge. In particular, if $d^+(v) < k$ for all vertices v , the algorithm will use at most k colours.

If we rewrite the proof of this fact (rather awkwardly) as a formal induction on k , the essential property of the set U of vertices coloured 1 is that every vertex in $G - U$ sends an edge to U : this ensures that $d_{G-U}^+(v) < d_G^+(v)$ for all $v \in G - U$, so we can colour $G - U$ with the remaining $k - 1$ colours by the induction hypothesis.

The following lemma generalizes these observations to list colouring, and to orientations D of G that do not necessarily come from a vertex enumeration but may contain some directed cycles. Let us call an independent set $U \subseteq V(D)$ a *kernel* of D if, for every vertex $v \in D - U$, there is an edge in D directed from v to a vertex in U . Note that kernels of non-empty directed graphs are themselves non-empty.

kernel

Lemma 5.4.3. *Let H be a graph and $(S_v)_{v \in V(H)}$ a family of lists. If H has an orientation D with $d^+(v) < |S_v|$ for every v , and such that every induced subgraph of D has a kernel, then H can be coloured from the lists S_v .*

Proof. We apply induction on $|H|$. For $|H| = 0$ we take the empty colouring. For the induction step, let $|H| > 0$. Let α be a colour occurring in one of the lists S_v , and let D be an orientation of H as stated. The vertices v with $\alpha \in S_v$ span a non-empty subgraph D' in D ; by assumption, D' has a kernel $U \neq \emptyset$.

α

D'
 U

Let us colour the vertices in U with α , and remove α from the lists of all the other vertices of D' . Since each of those vertices sends an edge to U , the modified lists S'_v for $v \in D - U$ again satisfy the condition

$d^+(v) < |S'_v|$ in $D - U$. Since $D - U$ is an orientation of $H - U$, we can thus colour $H - U$ from those lists by the induction hypothesis. As none of these lists contains α , this extends our colouring $U \rightarrow \{\alpha\}$ to the desired list colouring of H . $\square$

In our proof of the list colouring conjecture for bipartite graphs we shall apply Lemma 5.4.3 only to colourings from lists of uniform length k . However, note that keeping list lengths variable is essential for the proof of the lemma itself: its simple induction could not be performed with uniform list lengths.

Theorem 5.4.4. (Galvin 1995)

Every bipartite graph G satisfies $\text{ch}'(G) = \chi'(G)$.

(2.1.4) *Proof.* Let $G = (X \cup Y, E)$, where $\{X, Y\}$ is a vertex bipartition of G .
 X, Y, E We say that two edges of G meet in X if they share an end in X , and correspondingly for Y . Let $\chi'(G) = k$, and let c be a k -edge-colouring of G .
 k, c

H Clearly, $\text{ch}'(G) \geq k$; we prove that $\text{ch}'(G) \leq k$. Our plan is to use
 D Lemma 5.4.3 to show that the line graph H of G is k -choosable. To apply the lemma, it suffices to find an orientation D of H with $d^+(e) < k$ for every vertex e of H , and such that every induced subgraph of D has a kernel. To define D , consider adjacent $e, e' \in E$, say with $c(e) < c(e')$. If e and e' meet in X , we orient the edge $ee' \in H$ from e' towards e ; if e and e' meet in Y , we orient it from e to e' (Fig 5.4.3).

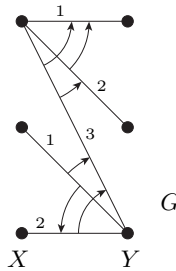


Fig. 5.4.3. Orienting the line graph of G

Let us compute $d^+(e)$ for given $e \in E = V(D)$. If $c(e) = i$, say, then every $e' \in N^+(e)$ meeting e in X has its colour in $\{1, \dots, i-1\}$, and every $e' \in N^+(e)$ meeting e in Y has its colour in $\{i+1, \dots, k\}$. As any two neighbours e' of e meeting e either both in X or both in Y are themselves adjacent and hence coloured differently, this implies $d^+(e) < k$ as desired.

D' It remains to show that every induced subgraph D' of D has a kernel. This, however, is immediate by the stable marriage theorem (2.1.4) for G , if we interpret the directions in D as expressing preference. Indeed, given a vertex $v \in X \cup Y$ and edges $e, e' \in V(D')$ at v , write $e <_v e'$ if the edge

ee' of H is directed from e to e' in D . Then any stable matching in the graph $(X \cup Y, V(D'))$ for this set of preferences is a kernel in D' . $\square$

By Proposition 5.3.1, we now know the exact list-chromatic index of bipartite graphs: (5.3.1)

Corollary 5.4.5. *Every bipartite graph G satisfies $\text{ch}'(G) = \Delta(G)$.* $\square$

5.5 Perfect graphs

As discussed in Section 5.2, a high chromatic number may occur as a purely global phenomenon: even when a graph has large girth, and thus locally looks like a tree, its chromatic number may be arbitrarily high. Since such ‘global dependence’ is obviously difficult to deal with, one may become interested in graphs where this phenomenon does not occur, i.e. whose chromatic number is high only when there is a local reason for it.

Before we make this precise, let us define two new invariants for a graph G . The greatest integer r such that $K^r \subseteq G$ is the *clique number* $\omega(G)$ of G . The greatest size of a set of independent vertices in G is the *independence number* $\alpha(G)$ of G . Clearly, $\alpha(G) = \omega(\overline{G})$ and $\omega(G) = \alpha(\overline{G})$. $\omega(G)$
 $\alpha(G)$

A graph is called *perfect* if every induced subgraph $H \subseteq G$ has chromatic number $\chi(H) = \omega(H)$, i.e. if the trivial lower bound of $\omega(H)$ colours always suffices to colour the vertices of H . Thus, while proving an assertion of the form $\chi(G) > k$ may in general be difficult, even in principle, for a given graph G , it can always be done for a perfect graph simply by exhibiting some K^{k+1} subgraph as a ‘certificate’ for non-colourability with k colours. *perfect*

At first glance, the structure of the class of perfect graphs appears somewhat contrived: although it is closed under induced subgraphs (if only by explicit definition), it is not closed under taking general subgraphs or supergraphs, let alone minors (examples?). However, perfection is an important notion in graph theory: the fact that several fundamental classes of graphs are perfect (as if by fluke) may serve as a superficial indication of this.²

What graphs, then, are perfect? Bipartite graphs are, for instance. Less trivially, the complements of bipartite graphs are perfect, too – a fact equivalent to König’s duality theorem 2.1.1 (Exercise 41). The

² The class of perfect graphs has duality properties with deep connections to optimization and complexity theory, which are far from understood. Theorem 5.5.6 shows the tip of an iceberg here; for more, the reader is referred to Lovász’s survey cited in the notes.